

Purchase Intent Classifier

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Large Language Models

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Final Project

Track A

The problem with traditional lead scoring

B2B companies receive hundreds of inbound leads per day via contact sales forms

Lead scoring helps classify leads by demographic traits and behavioral traits

In practice, lead scoring is somewhat weak in predicting a lead's intent to purchase

If a lead submits a contact sales request through the web form, the lead is automatically considered a **`hot`** lead by default, regardless of what it said

Sales wastes its time on leads with low intent to purchase but who consume a lot of marketing content

This lead would have a high lead score but would show a low intent to purchase, resulting in sales wasting its time if classified incorrectly

Sales misses high-intent leads who don't consume marketing content but who have a real need to purchase

This lead would have a low lead score but have a real intent to purchase, resulting in potential missed revenue if classified incorrectly

The gap

Traditional lead scoring asks...

Did the lead fill out
the form?

vs.

This LLM application asks...

What did the lead
actually say?

Project assumptions

- The name of my hypothetical company is **PipelineIQ** – we sell a CRM solution.
- I will model a **B2B enterprise technology company's** product, target audience, sales organization and sales process to set the stage for this application.
- This application is intended to **support the lead triage team** whose responsibility is to route leads to the right sales team.
- The dataset is built as if there is an **API integration between a CRM and the LLM application.**

Purchase Intent Classifier

A Jupyter notebook application that:

1. Receives inbound lead requests (contact sales message + demographic + behavioral data)
2. Classifies each lead across 3 dimensions using **GPT-4o**:
 - a. `intent_level` — high / medium / low / na
 - b. `opportunity_size` — large / medium / small / na
 - c. `purchase_timeline` — immediate / 3-6 months / 6+ months / na
3. Returns structured JSON output with classification + reasoning

What the LLM is evaluating

"We're currently using Salesforce but our team has been struggling with adoption and reporting. I'm evaluating alternatives for a potential Q1 rollout across our 200-person sales org. Can someone reach out this week to schedule a demo?"

"We've been evaluating CRM platforms for the past two months and PipelineIQ is on our shortlist. We need a solution for 500+ users with Salesforce integration. Can we schedule a call this week to discuss enterprise terms?"

"Our current CRM contract expires in 45 days and we need to migrate fast. We have 350 reps across North America. Please have someone contact me urgently — we're ready to move forward pending a final demo."

"We're consolidating three separate CRM tools into one platform for our global sales team of 400. Budget is approved. Need to see a demo this week and get a proposal by Friday."

Lead dataset (50 leads)

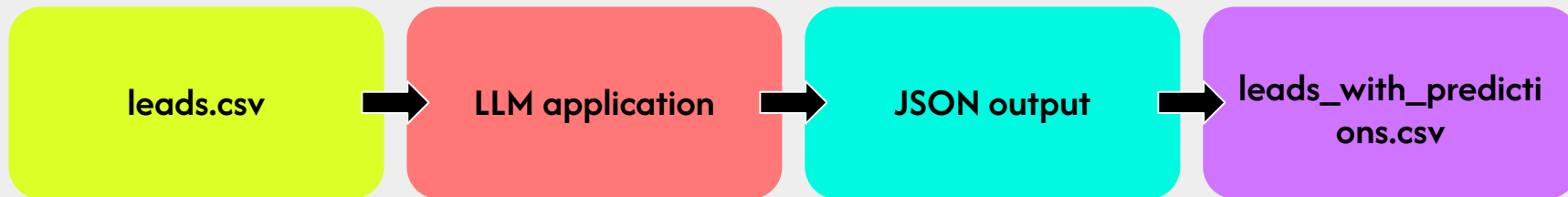
- The dataset consists of **50 leads** (17 columns) and are saved in leads.csv.
 - Demographic data
 - Behavioral data
 - Message submitted through the web form
 - Labels
- I used **Claude** to create the dataset and to label these leads across all three labels.
- **GPT-4o** will be classifying `intent_level`, `opportunity_size` and `purchase_timeline` based on **all of the columns** in the dataset (except the labels).
- **GPT-4o** will be asked to weight the message column most heavily, in addition to evaluating demographic and behavioral columns.
- For each lead, **GPT-4o** will also be providing its reasoning in the output.

Data flow

Theoretical production data flow:



Actual data flow for this project:



Prompt structure for each lead

System prompt

You are a helpful assistant helping a B2B sales triage team for a company that sells a leading CRM software. Your job is to analyze inbound lead submissions and classify each lead across three dimensions to help the triage team prioritize and route leads effectively.

Definitions

Rules

Few-shot examples

+

User prompt

Please classify the following inbound lead.

Contact ID: L018

Company: Apex Financial

Job Title: Director of Revenue Operations

Company Size: 201-1000 employees

Industry: Financial Services

Country: USA

Website Visits: 11

Content Downloads: 2

Attended Webinar: TRUE

Message: Just wanted to get some more information about your platform.

Example output

Purchase
Intent
Classifier

```
{  
  "intent_level": "medium",  
  "opportunity_size": "large",  
  "purchase_timeline": "3-6 months",  
  "reasoning": "The message is extremely vague,  
but strong behavioral signals indicate sustained  
engagement. Classified as large opportunity given  
Director-level title at a 201-1000 employee firm."  
}
```

Whereas traditional lead scoring rules...

Lead contacted sales → lead_score_level = hot ... when intent_level = medium

We'll run 3 prompts for comparison

Prompt Version Comparison

Prompt Version	Has Definitions	Has Rules	Has Examples
SYSTEM_PROMPT_V1	Yes	Yes	Yes
SYSTEM_PROMPT_V2	Yes	Yes	No
SYSTEM_PROMPT_V3	Yes	No	No

- The cost of running SYSTEM_PROMPT_V1 (**\$.005/per lead for ~1400 tokens**)
- At scale, the difference is ~\$200 to \$300 more to run SYSTEM_PROMPT_V1 per 100,000 leads
- Depending on performance, we will determine if this difference outweighs accuracy of the model

Prompt Version	Total Tokens / Lead	Est. Cost / Lead
SYSTEM_PROMPT_V1 (full)	1417	\$0.00504
SYSTEM_PROMPT_V2 (no examples)	639	\$0.00310
SYSTEM_PROMPT_V3 (bare minimum)	249	\$0.00212

Evaluation – Results

SYSTEM_PROMPT_V1 performs the best on all runs

Prompt Version	Tokens	intent_level	opportunity_size	purchase_timeline	Macro Avg
V1 (full)	~1400	0.8	0.98	0.9	0.893
V2 (no examples)	~600	0.72	0.9	0.88	0.833
V3 (bare minimum)	~250	0.72	0.86	0.74	0.773

- We want our application to be reliable; therefore, we consider the cost vs. accuracy and determine that improvement in accuracy outweighs the difference in cost
 - \$200-300 more per 100,000 leads
 - Macro-averaged accuracy score of ~89.3% across all runs (vs. ~83% and ~77%)
- Based on **SYSTEM_PROMPT_V1**'s performance, we choose to go with this prompt in the application
- We will focus the rest of the analysis on the performance of **SYSTEM_PROMPT_V1**

Evaluation – Metrics

Metric	Definition	Why It Matters for This Task
Accuracy	% of total predictions that are correct	An overall metric measuring how often the classifier is correct
Precision	Of all leads predicted as X, what % actually are X	Higher precision means the sales team is not wasting time following up on leads incorrectly predicted as higher intent
Recall	Of all leads that are truly X, what % did the model correctly identify	Higher recall ensures that valuable high-intent leads are not misclassified by the model as lower intent
F1 Score	Harmonic mean of precision and recall	Balances precision and recall into a single score
Macro-Averaged Accuracy	Mean accuracy across all three classification labels (<code>intent_level</code> , <code>opportunity_size</code> , <code>purchase_timeline</code>)	Summarizes overall model performance across all three classification tasks

SYSTEM_PROMPT_V1	Accuracy	Precision	Recall	F1
<code>intent_level</code>	80%	81%	79%	80%
<code>opportunity_size</code>	98%	98%	98%	98%
<code>purchase_timeline</code>	90%	91%	90%	90%

Conclusion – LLM performance

- ~90% macro-averaged accuracy across 3 classification tasks on 50 leads
- Strongest on opportunity_size (98%), most challenging on intent_level (84%)
- No geographic or gender bias detected; hierarchy sensitivity is expected but arguably reasonable behavior

Conclusion – Business impact

Traditional lead scoring

50 leads (100%) classified as “hot”
(equivalent to intent_level = high)

vs.

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19 of “hot” leads classified as
intent_level = high

15 (medium) + 16 (low) = 31 of
“hot” leads classified as intent_level
= medium or low

Estimated impact

62% of leads re-classified correctly
in terms of intent_level (vs.
classifying every lead who contacts
sales as “hot”)

Less time wasted

Potentially less revenue missed (vice
versa case)

This means sales teams can prioritize the top ~40% of **true high-intent leads** instead of chasing all 100% with the same amount of effort.*

Conclusion – Future work and next steps

Integrating this application to a real CRM instance like Hubspot or Pipedrive and automating lead enrichment using an LLM.

Analyzing real inbound lead submissions that a company like Hubspot might receive for their own CRM solution, which would provide a more reliable estimate of the model's performance with real data.

Expanding the dataset to 1000+ records to improve the robustness of the evaluation metrics that a 50-record database may not capture.

Adding a confidence score for each classification to the JSON output, so that the triage team can flag low-confidence predictions and manually review these leads accordingly.

With a large dataset of real labeled leads, we could also fine-tune a smaller model specifically on the lead classification task to improve accuracy on company-specific patterns, target audiences and behavioral traits that result in conversions.

Thank you!

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